

F1 Race Strategy Advisor

“Conditional strategy choice depends on the target: the one-stop M-H plan protects Top 10 probability, while the two-stop M-H-S plan improves Top 5 upside in the approved Dutch GP Scenario.”

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Repo: <https://github.com/Gottch7897/iit414w-lab01-TheUltrakills.git>

Commit: “[CAPSTONE] final-v1”

1. Executive Summary

The advisor supports a race strategy engineer making a pre-race call after qualifying. It compares one-stop and two-stop plans in a fixed driver-race context, reporting calibrated probabilities for points security (is_top10) and stronger upside (is_top5).

In the Dutch GP scenario, the one-stop protects Top 10 probability, while the two-stop improves Top 5 probability.

The recommendation is conditional, not automatic. The locked is_top10 model has Brier 0.139039 and ROC-AUC 0.881266, close to but **below the docent reference** of Brier 0.132 and ROC-AUC 0.892. The report therefore treats the model as useful decision support, not as a deployed strategy oracle.

2. Problem Framing

The supported decision is a pre-race strategy call before lights out, after qualifying information is available. The decision-maker is a **race strategy engineer** comparing whether a selected driver-race context should prioritize a one-stop or two-stop plan.

The prediction unit is one driver in one race. The primary target is “is_top10”; with an expansion target of “is_top5”. Strategy fields are scenario inputs, not ordinary pre-race predictors. **They would be leakage if used as observed post-race facts**, but here they are user-controlled what-if values.

3. Data and Validation

The race-level dataset has 2,447 driver-race rows from 2019-2024. The locked temporal split is train 2019-2021, calibration 2022, and final test 2023-2024. The **feature audit** uses pre-race context plus scenario inputs and excludes outcomes, realized weather, race incidents, DNF, and status columns from prediction.

Known limitations include:

- Short historical coverage
- Empty qualifying_time_s
- Coarse safety-car indicators
- Strategy choice, confounding with car pace, driver quality, traffic, weather, and race incidents.

4. Modeling Approach

The final narrative uses **separate logistic-regression models per target**: numeric features are imputed and standardized; categorical features are imputed and one-hot encoded.

Both targets use **sigmoid/Platt calibration** on 2022, selected because the calibration block is small and a smooth calibrator is less likely to overfit than isotonic calibration.

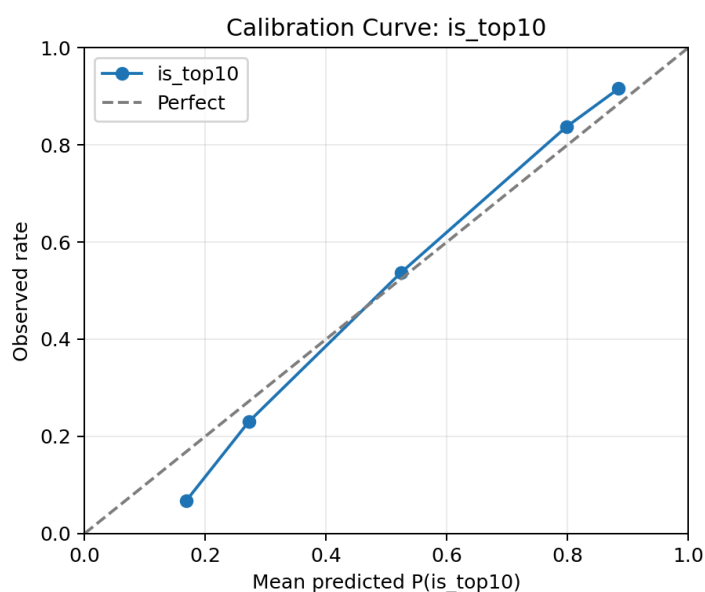
The model family is **conservative and explainable**. The cost is *possible underfitting*, which helps explain why the Top 10 model is close to but below the decent reference.

5. Results and Honest Comparison

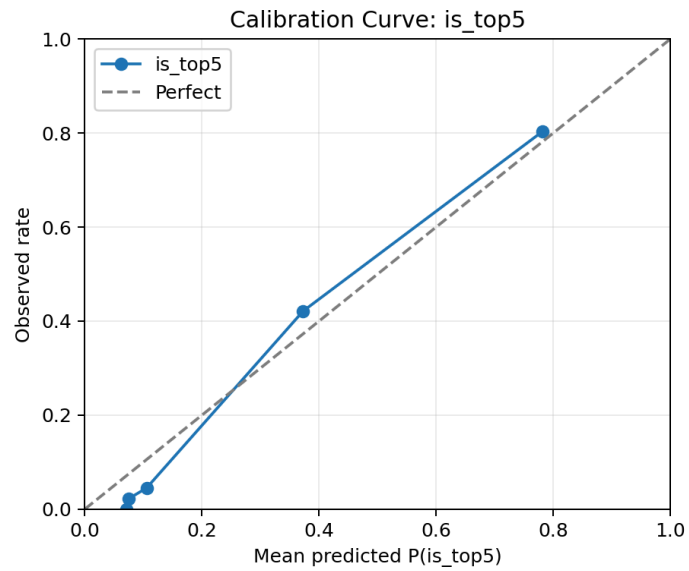
Target	Approach	Brier	Log Loss	ROC-AUC
'is_top10'	Target-rate baseline	0.249702	0.692551	0.500
'is_top10'	Calibrated logistic regression	0.139039	0.443436	0.881266
'is_top5'	Target-rate baseline	0.571728	0.571728	0.500
'is_top5'	Calibrated logistic regression	0.090479	0.305743	0.934413
<i>'is_top10'</i>	<i>Docent calibrated reference</i>	<i>0.132</i>	<i>Not Provided</i>	<i>0.892</i>

The Top 10 model improves strongly over the target-rate baseline but **does not beat the docent floor**. The Top 5 model has Brier 0.090479 and ROC-AUC 0.934413, which **supports the expansion target as a useful decision lens**.

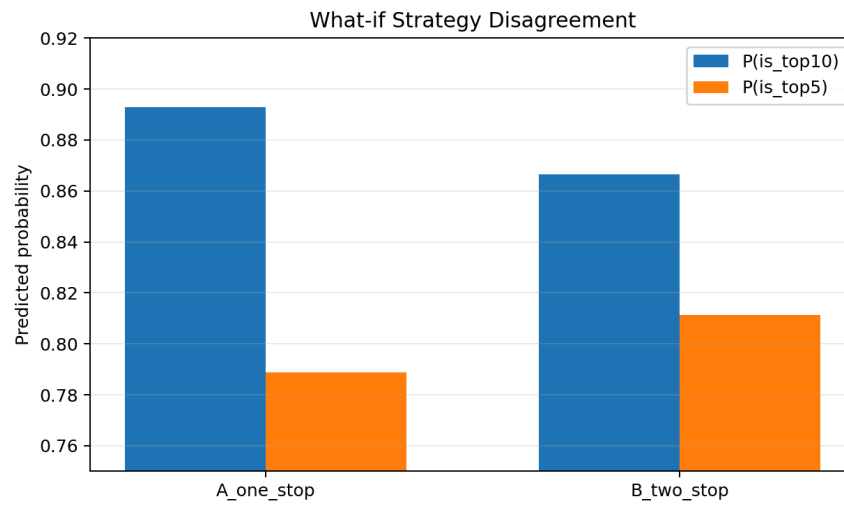
Calibration Evidence: is_top10



Calibration Evidence: is_top5



What-if Disagreement



6. Error Analysis and What-if

Target: Top10

Strategy Type	Rows	Actual Top 10 Rate	Mean Predicted Probability	Slice Brier	Mean Absolute Error
'three_plus_stop'	153	0.405	0.482	0.157	0.326
'one_stop'	353	0.561	0.580	0.147	0.314
'two_stop'	368	0.543	0.516	0.128	0.297
'no_stop'	15	0.000	0.145	0.022	0.145

Failure hypothesis: the model is weakest for "three_plus_stop" strategies on "is_top10". The mean prediction is higher than the actual rate by about 7.7 percentage points, suggesting that the model may not fully capture why extra stops happen. In real races, three-plus-stop strategies often reflect disruption, damage, poor tyre degradation, or recovery attempts, not only a planned aggressive strategy.

Circuit Type	Rows	Actual Top 10 Rate	Mean Predicted Probability	Slice Brier	Mean Absolute Error
'semi-street'	118	0.508	0.518	0.171	0.339
'permanent'	579	0.518	0.529	0.135	0.303
'street'	192	0.521	0.539	0.131	0.296

Failure hypothesis: "semi-street" circuits have the highest "is_top10" Brier score despite a moderate sample size of 118 rows. These races may combine street-circuit disruption with permanent-circuit pace patterns, making the simple circuit-type category too coarse for reliable probability estimates.

Constructor Tier	Rows	Actual Top 10 Rate	Mean Predicted Probability	Slice Brier	Mean Absolute Error
'midfield'	407	0.619	0.625	0.159	0.336
'backmarker'	312	0.189	0.249	0.132	0.305
'front'	170	0.876	0.814	0.105	0.236

Failure hypothesis: the largest "is_top10" error is for "midfield" constructors. This is plausible because midfield teams are exactly where strategy, traffic, and small pace differences most often decide whether a driver reaches the points. The model's average probability is close to the actual rate, but the high absolute error suggests it may be wrong on individual midfield cases even if the slice average looks calibrated.

Target: top5

Strategy Type	Rows	Actual Top 10 Rate	Mean Predicted Probability	Slice Brier	Mean Absolute Error
'three_plus_stop'	153	0.196	0.225	0.103	0.218
'one_stop'	353	0.283	0.314	0.097	0.224
'two-stop'	368	0.272	0.281	0.083	0.203
'no-stop'	15	0.000	0.088	0.012	0.088

Failure hypothesis: "three_plus_stop" remains the highest-Brier strategy slice for "is_top5", while "one_stop" has the highest mean absolute error. This suggests the model struggles to distinguish planned upside strategies from strategies caused by race disruption, especially when predicting stronger finishes rather than just points finishes.

Circuit Type	Rows	Actual Top 10 Rate	Mean Predicted Probability	Slice Brier	Mean Absolute Error
‘street’	192	0.260	0.283	0.099	0.224
‘permanent’	579	0.259	0.290	0.093	0.213
‘semi-street’	118	0.254	0.285	0.065	0.186

Failure hypothesis: for "is_top5", street circuits are the weakest circuit-type slice. A likely reason is that track position, safety-car timing, and overtaking difficulty matter more for Top 5 upside on street circuits, but the model only sees coarse circuit type and strategy categories.

Constructor Tier	Rows	Actual Top 10 Rate	Mean Predicted Probability	Slice Brier	Mean Absolute Error
‘front’	170	0.676	0.646	0.181	0.347
‘midfield’	407	0.278	0.287	0.114	0.257
‘backtracker’	312	0.006	0.075	0.011	0.080

Failure hypothesis: the "front" constructor tier is the highest-error slice for "is_top5". This is different from "is_top10", where front teams were the easiest slice. The difference makes sense: front cars almost always have high Top 10 probability, but Top 5 depends more on exact race execution, teammate competition, tyre timing, and whether the strategy converts pace into a high finish.

The main failure modes between the 2 targets are **midfield Top 10 predictions**, **three-plus-stop Top 10 strategies**, and **front-team Top 5 predictions**. However, these are operational warnings, not reasons to hide the Model.

For the strategy advisor, the practical consequence is that **a recommendation should not rely only on the overall model score**. If the proposed scenario involves a midfield constructor, a three-plus-stop strategy, or a Top 5 objective for a front team, the recommendation should include a reliability warning.

7. Limitations and Risks

The strongest limitation is **scenario dependence** under regime shift and observational confounding. Strategy choice is not random, so the what-if output is **not causal proof**. Single-team deployment is also risky because the selected scenario is midfield, and midfield Top 10 reliability is the weakest constructor-tier slice.

We do not recommend deploying this tool, unless:

1. The strategy desk **limits** it to pre-race what-if comparisons with fixed driver-race context,
2. Every recommendation displays **target-specific slice reliability** for strategy type, circuit type, and constructor tier, and
3. The team **validates the model** on a fresh race weekend or simulator backtest.

8. Reproducibility and AI Reflection

Run `venv/bin/python Capstone/Final_Report/generate_final_report.py` from the repository root to regenerate the tables, figures and Markdown report.

The PDF itself was made from the Markdown report and manually revised by the team, confirming that the data matched with the correct Markdowns.

- The Hito 1 and Hito 2 source artifacts remain in Capstone/ and all model `random_state` arguments use `RANDOM_SEED = 414`.
- AI assistance was used for structure, prose, captions, critique, and Q&A drafting.
- Outputs were validated against the locked Hito evidence.
- Suggestions that overstated causality or claimed the Top 10 model beat the docent reference were rejected.

9. References

- IIT414W course capstone brief and Canvas rubrics. (2026). F1 Race Strategy Advisor.
- Course dataset: fl_strategy_race_level.csv, seasons 2019-2024.
- FastF1 project documentation and historical Formula 1 timing data sources.
- Joplica F1 API documentation and historical race metadata sources.
- scikit-learn developers. (2026). Logistic regression, calibration, and model evaluation/documentation.